

Customer Shopping Behavior Analysis

1. Project Overview

This project analyzes customer shopping behavior using a retail transaction dataset containing **3,900 customer records**. The objective is to identify customer purchasing patterns, product performance, and revenue trends through data analysis. The project includes **data cleaning and exploratory data analysis (EDA) in Python**, **business analysis using MySQL**, and the development of an **interactive Power BI dashboard**. The insights obtained can help businesses improve marketing strategies, customer engagement, and inventory planning.

2. Dataset Summary

The dataset used in this project contains **3,900 customer shopping records** with **18 columns**, where each row represents an individual customer purchase. It includes customer demographics, product information, purchase details, payment methods, shipping preferences, customer reviews, discounts, and shopping behavior. During the data cleaning process, **37 missing values** were identified in the **Review Rating** column and were handled using **category-wise median imputation** to maintain data quality.

Dataset Features

- **Total Records:** 3,900
- **Total Columns:** 18
- **Customer Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Item Purchased, Category, Purchase Amount, Season, Size, Color
- **Shopping Behavior:** Discount Applied, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
- **Missing Data:** 37 missing values in the **Review Rating** column (handled during preprocessing)

3. Exploratory Data Analysis using Python

We began with data preparation and exploratory data analysis using **Python (Pandas)** to ensure the dataset was clean, consistent, and ready for further analysis.

- **Data Loading:** Imported the dataset using the **Pandas** library and loaded it into a **DataFrame**.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

- **Initial Exploration:** Used `df.info()`, `df.describe()`, and `df.isnull().sum()` to examine the dataset structure, data types, summary statistics, and missing values.

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3900 entries, 0 to 3899
Data columns (total 18 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Customer ID                          3900 non-null   int64
1   Age                                  3900 non-null   int64
2   Gender                              3900 non-null   object
3   Item Purchased                       3900 non-null   object
4   Category                             3900 non-null   object
5   Purchase Amount (USD)                3900 non-null   int64
6   Location                             3900 non-null   object
7   Size                                 3900 non-null   object
8   Color                                3900 non-null   object
9   Season                               3900 non-null   object
10  Review Rating                        3863 non-null   float64
11  Subscription Status                  3900 non-null   object
12  Shipping Type                       3900 non-null   object
13  Discount Applied                    3900 non-null   object
14  Promo Code Used                     3900 non-null   object
15  Previous Purchases                  3900 non-null   int64
16  Payment Method                      3900 non-null   object
17  Frequency of Purchases              3900 non-null   object
dtypes: float64(1), int64(4), object(13)
memory usage: 548.6+ KB

```

```

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    37
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64

```

- **Missing Data Handling:** Identified **37 missing values** in the **Review Rating** column and filled them using the **median review rating of each product category**, preserving category-specific trends. Used median to fill the missing values since it's more robust to outliers.

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
```

```
df.isnull().sum()
```

```

Customer ID      0
Age              0
Gender           0
Item Purchased   0
Category         0
Purchase Amount (USD)  0
Location         0
Size            0
Color           0
Season          0
Review Rating    0
Subscription Status  0
Shipping Type    0
Discount Applied  0
Promo Code Used  0
Previous Purchases  0
Payment Method   0
Frequency of Purchases  0
dtype: int64

```

- **Column Standardization:** Renamed all column names to **snake_case** to improve readability, maintain consistency, and simplify SQL querying.

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'promo_code_used', 'previous_purchases',  
      'payment_method', 'frequency_of_purchases'],  
      dtype='object')
```

- **Feature Engineering:** Created a new **age_group** column by grouping customers into different age categories to support demographic analysis.

	age	age_group
0	55	Middle-aged
1	19	Young adult
2	50	Middle-aged
3	21	Young adult
4	45	Middle-aged
5	46	Middle-aged
6	63	Senior
7	27	Young adult
8	26	Young adult
9	57	Middle-aged

- **Data Consistency Check:** Verified that the **promo_code_used** column contained redundant information with **discount_applied** and removed it to eliminate unnecessary duplication.

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',  
      'purchase_amount', 'location', 'size', 'color', 'season',  
      'review_rating', 'subscription_status', 'shipping_type',  
      'discount_applied', 'previous_purchases', 'payment_method',  
      'frequency_of_purchases', 'age_group', 'purchase_frequency_days'],  
      dtype='object')
```

- **Export Cleaned Dataset:** Saved the cleaned dataset as a CSV file and imported it into **MySQL** for SQL-based business analysis. The cleaned dataset was also used as the data source for the **Power BI dashboard**.

4. Data Analysis using SQL (Business Transactions)

After cleaning the dataset, it was imported into **MySQL** to perform business-oriented analysis. The following SQL queries were executed to extract meaningful business insights from the customer shopping dataset:

1. Revenue by Product Category

Objective: Identified the product categories generating the highest total revenue.

	category	revenue_by_category
►	Clothing	104264
	Footwear	36093
	Outerwear	18524
	Accessories	74200

2. Subscribers vs. Non-Subscribers Spending

Objective: Compared average purchase amount and total revenue between subscribed and non-subscribed customers.

	avg_purchase_amount	total_revenue	subscription_status
►	59.4919	62645	Yes
	59.8651	170436	No

3. Highest-Rated Products

Objective: Identified the top 5 products with the highest average customer review ratings.

	item_purchased	avg_review_rating
►	Gloves	3.86
	Sandals	3.84
	Boots	3.82
	Hat	3.8
	Handbag	3.78

4. Top 3 Products within Each Category (*Window Function – DENSE_RANK*)

Objective: Ranked the three most preferred products within every product category.

	item_purchased	category	total_purchases	rank_num
▶	Jewelry	Accessories	171	1
	Sunglasses	Accessories	161	2
	Belt	Accessories	161	2
	Scarf	Accessories	157	3
	Blouse	Clothing	171	1
	Pants	Clothing	171	1
	Shirt	Clothing	169	2
	Dress	Clothing	166	3
	Sandals	Footwear	160	1
	Shoes	Footwear	150	2
	Sneakers	Footwear	145	3
	Jacket	Outerwear	163	1
	Coat	Outerwear	161	2

5. Products with High Sales but Low Ratings (*CTE + Subquery*)

Objective: Identified products that generate strong sales despite receiving below-average customer ratings.

	item_purchased	avg_review_rating	total_sales
▶	Blouse	3.68	10410
	Shirt	3.62	10332
	Shorts	3.71	9433
	Dress	3.75	10320
	Sunglasses	3.75	9649
	Pants	3.72	10090
	Scarf	3.71	9561

6. Revenue by Customer Segment (*CASE Statement*)

Objective: Compared revenue generated by New, Returning, and Loyal customers.

	revenue	customer_segment
▶	185517	Loyal
	42711	Returning
	4853	New

7. Top 5 Customers by Total Spending (*ROW_NUMBER*)

Objective: Ranked the highest-value customers based on their total purchase amount.

	customer_id	total_amount	rank_num
▶	43	100	1
	96	100	2
	194	100	3
	205	100	4
	244	100	5

8. Above-Average Spending Customers (*CTE + Subquery*)

Objective: Identified customers whose total spending exceeds the overall average customer spending.

	customer_id	total_amount
▶	2	64
	3	73
	4	90
	7	85
	9	97
	12	68
	13	72
	16	81
	20	90
	22	62
	24	88
	29	94
	32	79
	33	67
	35	91
	37	69
	40	60
	41	76
	43	100

9. Category-wise Revenue Contribution (*Window Function*)

Objective: Calculated the percentage contribution of each product category to the overall revenue.

	category	percentage_contribution
▶	Clothing	44.73
	Footwear	15.49
	Outerwear	7.95
	Accessories	31.83

10. Top-performing Product by Age Group (*DENSE_RANK*)

Objective: Identified the highest revenue-generating product within each customer age group.

	age_group	item_purchased	revenue	rank_num
►	Adult	Blouse	3240	1
	Middle-aged	Shorts	3138	1
	Senior	Jewelry	3107	1
	Young adult	Shirt	3343	1

11. Customer Spending Tier Classification (*CASE Statement*)

Objective: Classified customers into Bronze, Silver, Gold, and Platinum tiers based on their total purchase amount.

	customer_id	total_amount	spending_tier
	2995	100	Silver
	3029	100	Silver
	3266	100	Silver
	3322	100	Silver
	3726	100	Silver
	3838	100	Silver
	92	99	Bronze
	293	99	Bronze
	414	99	Bronze
	458	99	Bronze
	500	99	Bronze
	564	99	Bronze

12. Revenue by Gender

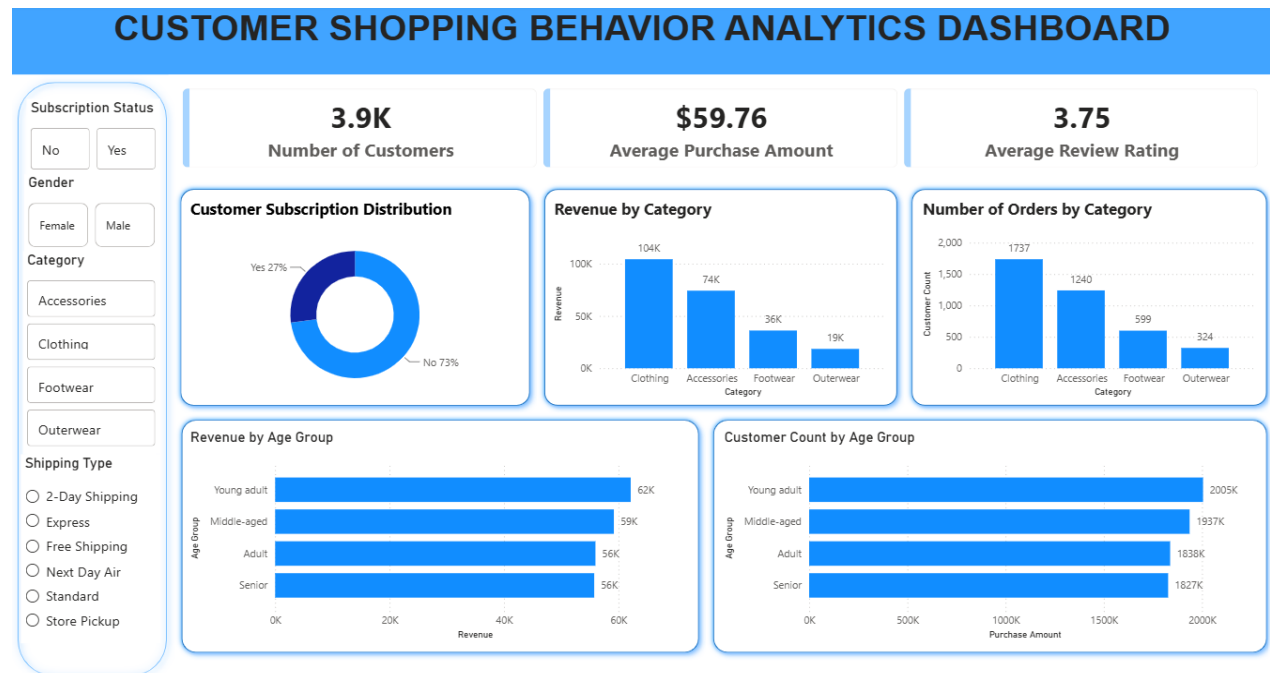
Objective: Analyze revenue contribution by gender to support customer segmentation and targeted marketing strategies.

	gender	total_revenue
►	Male	157890
	Female	75191

5. Dashboard in Power BI

An interactive dashboard was developed using **Power BI** to visualize key business metrics and customer shopping trends. The dashboard includes KPIs, interactive slicers, and visualizations for revenue analysis, customer segmentation, product performance, and purchase behavior.

It enables users to explore the data dynamically and gain actionable business insights for informed decision-making.



6. Business Recommendations

- **Strengthen Customer Subscription Programs** – Enhance subscription benefits and promotional offers to encourage more customers to join and improve long-term customer retention.
- **Focus on High-Revenue Product Categories** – Increase inventory, marketing efforts, and promotional campaigns for categories contributing the highest revenue.
- **Reward High-Value Customers** – Introduce loyalty programs and exclusive rewards for Gold and Platinum customers to improve customer retention and lifetime value.
- **Promote Top-Rated and Best-Selling Products** – Highlight highly rated and frequently purchased products in marketing campaigns to maximize sales and customer satisfaction.
- **Optimize Customer Segmentation Strategies** – Design personalized marketing campaigns based on customer segments, age groups, and gender to improve engagement and conversion rates.
- **Review Underperforming Products** – Investigate products with high sales but lower customer ratings to identify quality improvements and enhance customer satisfaction.

7. Conclusion

This project successfully analyzed customer shopping behavior using Python, MySQL, and Power BI. Business insights were generated through data cleaning, SQL-based analysis, and

interactive dashboard development. The findings can support better decision-making in customer segmentation, inventory planning, and marketing strategies. The project also demonstrates practical data analytics skills across the complete analytics workflow.